

In []:

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In [1]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
import matplotlib.pyplot as plt

url="https://raw.githubusercontent.com/justmarkham/pycon-2016-tutorial/master/data.csv"
df=pd.read_csv(url,sep='\t',names=['label','message'])

print(f"Total: {len(df)} | Ham: {sum(df['label']=='ham')} | Spam:{sum(df['label']=='spam')}")

df['label_num']=df['label'].map({'ham':0,'spam':1})
X_train,X_test,y_train,y_test=train_test_split(
    df['message'],df['label_num'],test_size=0.2,random_state=42
)

tfidf=TfidfVectorizer(max_features=5000,stop_words='english')
X_train_vec=tfidf.fit_transform(X_train)
X_test_vec=tfidf.transform(X_test)

model=MultinomialNB()
model.fit(X_train_vec,y_train)

y_pred=model.predict(X_test_vec)
print(f"\nAccuracy:{(y_pred==y_test).mean():.2%}")

fig,axes=plt.subplots(1,2,figsize=(10,4))

df['label'].value_counts().plot(kind='pie',autopct='%1.1f%%',ax=axes[0])
axes[0].set_title('Ham vs Spam')

feature_names=tfidf.get_feature_names_out()
spam_score=model.feature_log_prob_[1,:]-model.feature_log_prob_[0,:]
top_idx=spam_score.argsort()[-10:]

axes[1].barh(range(10),spam_score[top_idx])
axes[1].set_yticks(range(10))
axes[1].set_yticklabels([feature_names[i]for i in top_idx])
axes[1].set_title('Top 10 Spam Words')

plt.tight_layout()
plt.show()

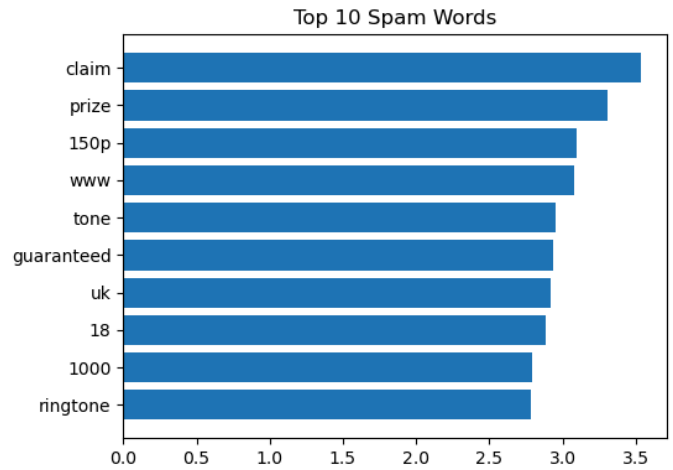
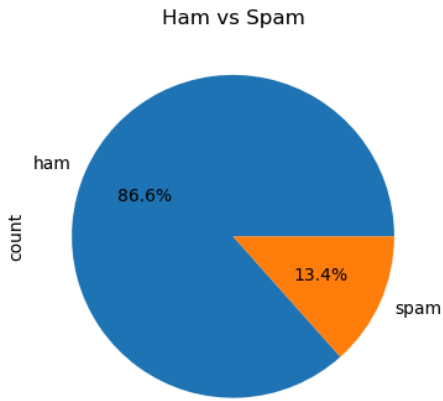
test_messages=[
    "Free prize!Claim now!",
    "Hey,metting at 3pm tomarrow",
    "You won $1000!Click here"
]

for msg in test_messages:
    vec=tfidf.transform([msg])
    pred=model.predict(vec)[0]
    print(f"\n'{msg[:30]}...'-'{ 'SPAM'if pred==1 else 'HAM'}")

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Total: 5572 | Ham: 4825 | Spam:0

Accuracy:97.94%



'Free prize!Claim now!...'-SPAM

'Hey,metting at 3pm tomarrow...'-HAM

'You won \$1000!Click here...'-SPAM

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